

LODE

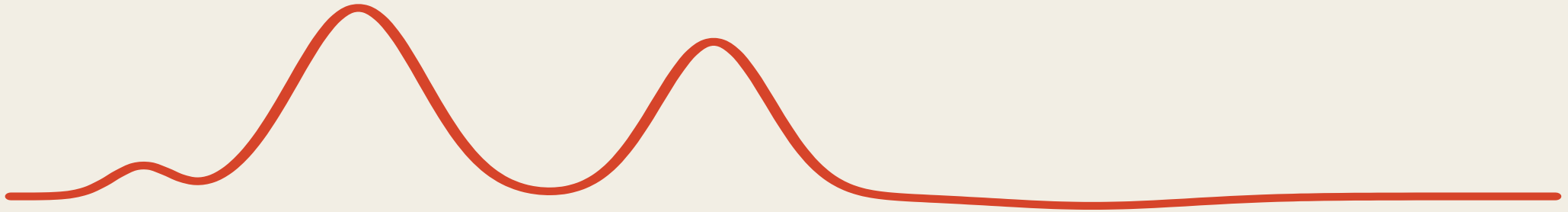
Proof of work where the scarce input is you.

CLOCK IN · SOLANA MOBILE · TEAM HER

THE THESIS

Your phone only mines while it can prove a real person is walking.

Walk longer and it's allowed to search harder. Stop, and the budget decays to zero inside a minute.



THE PROBLEM

Proof of work has exactly one input: electricity.

Whoever buys the most power wins. That's why mining left bedrooms and ended up in warehouses beside power stations, and why ordinary people stopped being able to take part years ago.

The same capture happens everywhere crypto tries to reach people:

MECHANISM	HOW IT GETS CAPTURED
Airdrops	One operator, a thousand wallets
Captchas	Solved by scripts for a fraction of a cent
Proof of personhood	Your passport, or a scan of your iris

WHY EFFORT, SPECIFICALLY

Hashrate is for sale. An hour of walking isn't.

Capital converts straight into electricity, into GPUs, into wallets, into solved captchas — scale is a purchase order. Physical effort is the one input that doesn't convert. Ten thousand GPUs is an afternoon and a credit card. Ten thousand people walking for an hour is ten thousand people walking for an hour.

INPUT	PRICE AT SCALE
Hashrate	Rentable by the hour
Wallets	Free, unlimited
Captcha solves	A tenth of a cent each
Proof-of-personhood bypass	Whatever an identity costs
An hour of walking	**—**

THE HARD PART

A step counter is trivially faked.

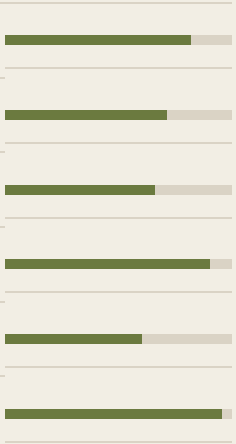
Shake your phone. Earn forever. If that works, none of the rest matters. So the entire product reduces to one question:

Can a phone tell the difference between a person walking and a person pretending?

SIX SIGNALS, FIFTY TIMES A SECOND

What the instrument reads.

CADENCE	Stride sits between 1.2 and 3.4 steps per second
STEP VAR	Gait repeats approximately — never exactly, never randomly
SKEW	Footfalls push acceleration up more sharply than down
KURTOSIS	Shaking spikes far harder than a footfall
ORIENT	A carried phone wanders; a mounted one never does
GYRO CORR	A walking body rotates as it steps



Computed on device over a five-second window, refreshed every second. Raw samples never persist and never leave the phone.

THE ONE THAT'S HARD TO FAKE

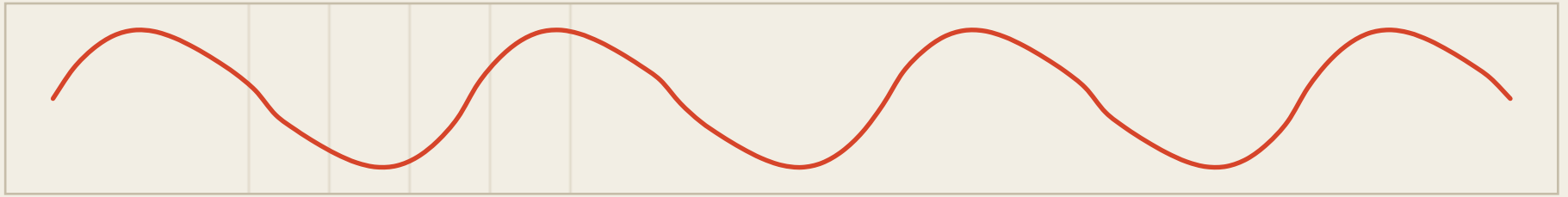
A walking body rotates as it steps.

A phone shaken in the hand translates without rotating. That coupling between rotation and vertical motion is the signature a faker has to reproduce — and reproducing it means actually moving like a person.

WALKING

0.89

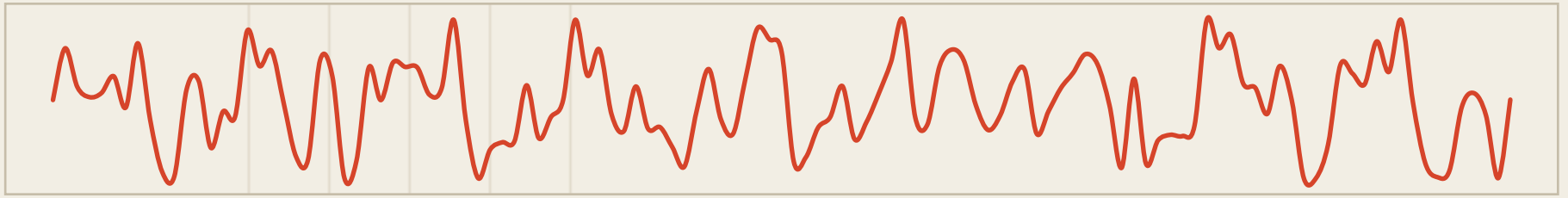
GYRO CORR



SHAKEN

0.11

GYRO CORR



WHY YOU CAN'T FAKE FIVE AND CARRY THE SIXTH

We combine the signals geometrically, not on average.

An average lets five soft passes outvote one hard failure. We found this the hard way: a shaken phone scored 0.74 and credited, while the only signal that genuinely rejected it was drowned out by the other five.

Combining them so the score collapses when any single signature goes missing changed that result to 0.299 — refused.

A motion floor sits ahead of all six: the features are shape statistics, and the shape of sensor noise lands inside the walking bands by accident. Without it, a phone sitting still on a desk scores 0.91.



THE SAME SHAKE, BEFORE AND AFTER

WHAT THE CHAIN ACTUALLY VERIFIES

It recomputes the work. It doesn't take our word for it.

An Anchor program on Solana devnet:

recomputes `sha256(challenge ‖ authority ‖ nonce)` itself

counts the leading zero bits and rejects anything below the claim

rejects a reused nonce, and rate-limits submissions by slot

mints scaled to the difficulty actually achieved

PROGRAM `ADYPeA3jNDXoXQjYSTXwLCXjFiWJWZx99ANRnhiGf5Wp`
MINT `8vRrdq81gjZVcCFFCrZcjd2AKrkenKmSw5ndhYcUmdrN`

22
DIFFICULTY FOUND

250,618
HASHES SEARCHED

1,638.40
LODE PAID

WHAT IT DOES NOT VERIFY

The chain can’t check the walking. We’re not going to pretend it can.

The program verifies the hash and the signature. It cannot verify the classification, because on-chain verification of sensor data isn’t possible. That runs on the device, and the program trusts it.

Hardware attestation — proving the binary is unmodified — closes that gap. It’s the next thing we build, and a device with a secure element is exactly where it belongs.

WHY THIS IS A MOBILE PROJECT

It could not exist anywhere else.

01 The input is motion hardware. There is no web version of this, and no desktop version — the sensors are the product.

02 Effort claims are signed through Mobile Wallet Adapter, which on a Seeker routes to the Seed Vault Wallet with no Lode-specific code path.

03 Seeker is also where the open problem closes: attestation needs a secure element, and that's the one piece of hardware that has it.

THE REFRAME

The mining is the demo. The gate is the product.

Proof that a human did something physical is a primitive — mining is simply the first thing we put behind it. Anything that needs to reach people who actually showed up runs into the same wall we just built a door in.

SCOPE, AND WHAT'S NEXT

What we deliberately left out.

Background sessions with the screen off · hardware attestation · multi-device accounts · leaderboards · anything social · iOS

Three days, one builder. A short deliberate scope is a decision. A long apologetic one is a list of things you ran out of time for.

Next: attestation on Seeker, then calibration against a wider range of gaits and carry positions.

REPO github.com/Lideeyah/lode

APK github.com/Lideeyah/lode/releases/latest

DEMO youtube.com/shorts/PFBdXa-BeTs

LODE